This article has been accepted for publication in IEEE Internet of Things Journal. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JIOT.2026.3741571

Rethinking the Reliability of Trajectory Privacy through Group Based Semantic Transition

Yude Bai, Zitian Liu, Lyuchuan Wei, Minhong Dong, Guangquan Xu, Xiaoqiang Di, Pan Gao*, Wei Yu*, Wenying He*

***Abstract*—Location-based services in intelligent IoT systems rely on trajectory data to support context-aware applications. In realistic urban environments, uneven Points-of-Interest distributions significantly amplify semantic imbalance and weaken conventional anonymization strategies. Semantic imbalance across geographic regions distorts transition regularities and introduces privacy vulnerabilities under Dynamic Semantic Probability Distribution Attacks. To this end, we propose GeoTrace, a geospatial-aware trajectory group semantic anonymization framework. GeoTrace explicitly incorporates semantic transition statistics into the anonymization process to regulate pseudo-location selection. It also integrates temporal, spatial, and semantic information through group-based deep semantic trees and entropy-guided pseudo-location selection. GeoTrace further constrains candidate selection through group-level semantic aggregation to expand the feasible anonymization space. This design mitigates sparsity-induced instability and preserves semantic diversity. Experiments on the Geolife dataset show that GeoTrace maintains Transfer Direction Deviation above 0.9 across different K values. GeoTrace keeps Top-n Position Obfuscation Attack success rates below 20%, indicating reduced attack effectiveness.**

***Index Terms*—Trajectory Privacy, Semantic Transition, Group Semantic Tree, Dynamic Semantic Probability Distribution Attack**

I. Introduction

As an integral component of IoT networks, Location-Based Services play a critical role in intelligent edge ecosystems. These systems rely on trajectory data generated by mobile users and devices to provide location-aware services, including next points-of-interest recommendations [1] and travel suggestions [2]. The rapid proliferation of such services, however, raises serious security and privacy concerns for mobile nodes. Adversaries can analyze trajectory patterns, query behaviors, and semantic transitions to infer sensitive information. Such inference risks threaten both user privacy and overall system reliability [3]. Therefore, trajectory anonymization is required to hide sensitive mobility patterns while preserving the contextual utility of trajectory data.

Yude Bai and Zitian Liu, School of Software, Tiangong University, Tianjin, China (Email: baiyude@tiangong.edu.cn; 2531111377@tiangong.edu.cn). Lyuchuan Wei, Minhong Dong and Guangquan Xu, School of Cybersecurity, Tianjin University, Tianjin, China (Email: wlc_aobo@tju.edu.cn; dongminhong@tiangong.edu.cn; losin@tju.edu.cn). Xiaoqiang Di, Jilin Province Key Laboratory of Network and Information Security, Changchun University of Science and Technology, Changchun, China (Email: dixiaoqiang@cust.edu.cn) Pan Gao, Shihezi University, Xinjiang, China (Email: gp_inf@shzu.edu.cn). Wei Yu, Zhejiang Yuexiu University, Zhejiang, China (Email: weiyu@zyufl.edu.cn). Wenying He, School of Artificial Intelligence, Hebei University of Technology, Tianjin, China. (Email: hwy1234@hebut.edu.cn).

Copyright (c) 2026 IEEE. Personal use of this material is permitted. However, permission to use this material for any other purposes must be obtained from the IEEE by sending a request to pubs-permissions@ieee.org.

This requirement becomes more stringent in edge-assisted IoT environments, where pseudo-locations must support real-time decisions [4]. Timely location updates support task scheduling, resource allocation, and service coordination [5], but semantically unmatched dummy trajectories may mislead edge servers and reduce decision reliability [6]. Anonymization should therefore increase location uncertainty while preserving semantic plausibility and mobility consistency.

Many privacy-preserving approaches focus on trajectory semantics and geographic transition mechanisms [7]. Semantic-aware methods use trajectory semantics to guide pseudo-location selection [8], such as MSP's personalized semantic trees [9], while preliminary group semantic anonymization (TGSA) consolidates individual semantic variances [10]. Others emphasize geographic transitions, such as the RDG model [11], [12]. These methods improve privacy under specific assumptions but remain insufficient for dynamic edge environments, personalized semantic structures depend on individual trajectory density and suffer from sparsity-induced instability, and geographic-transition methods assume that geographic continuity aligns with semantic plausibility [13]. Under uneven semantic distributions, however, geographic density and semantic transition likelihood can diverge, allowing attackers to exploit transition inconsistencies [9], [11]. Recent graph-based and LLM-based paradigms address release objectives that differ from conditional semantic inference within published anonymity sets.

Sparsity and imbalance describe different difficulties. Spatial sparsity means limited and unevenly distributed observations for an individual or region. It leaves transition counts incomplete and unstable and restricts the supportable semantic categories. Category imbalance means the conditional distribution of the next semantics given the current semantic is skewed. It concentrates probability mass on a few categories, lowers the conditional transition entropy, and makes the next semantic more predictable. Sparsity affects coverage and estimation reliability, whereas imbalance affects concentration and predictability.

A semantic transition pattern is the conditional distribution of the next semantic category $y = c_{q+1}$ given the current stay-point category c_q for user u. It is a distribution over candidates rather than a deterministic route. Hence a marginal entropy, or an entropy estimated from too few observations, does not characterize the uncertainty under conditional inference.

This limitation becomes more evident under uneven POI distributions [14]. Our analysis within the six-ring road of Beijing records 85 to 4,499 stay points across nine grids, demonstrating highly unequal spatial support, but this range

Authorized licensed use limited to: Tiangong University. Downloaded on October 12,2026 at 10:58:42 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Internet of Things Journal. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JIOT.2026.3741571

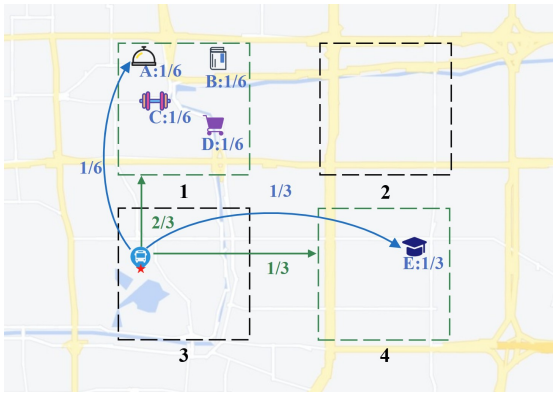


Fig. 1. A 4-grid example of uneven distribution of POI.

does not imply lower entropy. The risk arises because sparse grids offer limited semantic support with skewed or unstable transition counts. An attacker can then exploit conditional regularities even when candidates are geographically dispersed or marginal entropy is high, so group-level aggregation is needed to enlarge and stabilize semantic support [3], [14]–[17].

The 4-grid example in Fig. 1 illustrates this. A user departs from a station marked by a star. Geographic-transition defenses such as RDG [11] favor Grid 1 for its high geographic transition probability, while the semantic pattern from Station assigns $1/3$ to School, Point E in Grid 4, and $1/6$ to each of Restaurant, Library, Gym, and Shop, Points A–D in Grid 1. Point E thus has the highest conditional probability. This non-uniform candidate distribution has lower entropy than a uniform one, letting an attacker condition on the current semantic, rank K candidates by historical conditional probabilities, and discard low-probability dummies even when marginal entropy is high. Dynamic Semantic Probability Distribution Attacks exploit exactly this mismatch between geographic plausibility and semantic likelihood [4], [11].

To address this challenge, we propose Geospatial-Aware Trajectory Group Semantic Anonymization (GeoTrace). The Data Preparation module extracts stay points and partitions users into lifestyle groups by semantic transition similarity. The Group Deep Semantic Tree module aggregates shared semantic nodes across users with similar mobility behaviors. The Group Semantics Entropy Anonymization module selects pseudo-locations via entropy-guided selection and spatial constraints to produce K-anonymous sets.

Evaluation uses Transfer Direction Deviation, Average Identification Possibility, Semantic Transfer Entropy, and Top-n Position Obfuscation Attack Success Rate [4], [13], [18]. GeoTrace maintains DD above 0.9 across K values and keeps Top-n attack success rates below 20%, while achieving comparable AIP and Semantic Transfer Entropy to RDG and lower Top-n attack success rates than MSP.

The main contributions are summarized as follows.

- We propose GeoTrace, a trajectory privacy preservation framework, to address vulnerabilities induced by semantic imbalance. It considers temporal, spatial, and semantic factors to build privacy-preserving trajectory representations.
- A group-based deep semantic tree construction method is introduced to mitigate sparsity-induced instability in personalized semantic structures and improve semantic consistency across users with similar mobility patterns. This mechanism further enlarges the feasible pseudo-location candidate space while preserving semantic diversity, spatial plausibility, and privacy protection.
- Evaluation on the Geolife dataset shows high trajectory direction consistency, dispersed conditional transition distributions, and lower Top-n attack success rates than MSP.

II. Methodology

A. Formal Definitions

Table I summarizes the key notations and formal definitions used to describe user trajectories, stay point extraction, and semantic transition representations.

B. Trajectory Privacy Threats

Trajectory privacy preservation exhibits inherent vulnerabilities when semantic distributions remain uneven across geographic regions. Adversaries exploit inconsistencies between geographic plausibility and semantic transition regularities to reduce anonymity sets [4], [11], [14]. The following attacks represent typical inference risks that directly affect dummy location selection and anonymization reliability.

Single-point attack [8] exploits auxiliary information such as edge-side context or spatial constraints to filter out dummy locations and narrow the anonymity scope. Poorly chosen dummies that violate basic spatial plausibility, for example in lakes or inaccessible areas, can be detected and discarded before further inference.

Speed anomaly attack [12] targets anonymized trajectories with abrupt changes in speed or direction. Dummies that introduce sudden directional deviations or velocity mutations can be distinguished from genuine mobility patterns, weakening the credibility of the dummy set.

Static probability distribution attack [11] uses historical service request logs to estimate query probability distributions, exploiting concentrations produced by uneven semantic or geographic distributions.

Dynamic semantic probability distribution attack extends static strategies by incorporating semantic transition statistics rather than raw query frequencies. Semantic imbalance alters regularities in urban mobility. Attackers estimate the conditional distribution of the next semantic category from historical queries and, given the current category and K candidate locations, rank candidates by their estimated transition probabilities before eliminating low-probability pseudo-locations. Unlike static attacks based primarily on query frequencies, this attack exploits sequential semantic dependencies to progressively narrow the anonymity set. We introduce Semantic Transition Entropy to quantify the uncertainty that remains under this conditional inference. The semantic query probability $P(j)$ for semantic class j is defined as follows.

Authorized licensed use limited to: Tiangong University. Downloaded on October 12,2026 at 10:58:42 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Internet of Things Journal. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JIOT.2026.3741571

TABLE I
SUMMARY OF KEY NOTATIONS AND DESCRIPTIONS

Notation	Description
$L_s = \{ls_1^u, \ldots, ls_n^u\}$	Sequence of n check-ins for user $u \in U$, where each check-in is denoted by (l, t), with $l \in L$ and t as the access time.
$dur(l_x^u, l_y^u)$, $dis(l_x^u, l_y^u)$	The stay point set L_{stay} is formed when $dis(l_x^u, l_y^u) \leq \theta_d$ and $dur(l_x^u, l_y^u) \geq \theta_t$. Representative stay points are selected from this set.
$LS_{\text{stay}} = \{ls_1^u, \ldots, ls_n^u\}$	The trajectory of stay points for user u, consisting of a sequence of representative stay points.
M_C	The generalized semantic transition matrix computed for each user u based on their semantic sequence C_u.
M_P	The overall generalized semantic transition matrix obtained by aggregating M_C across all users.
$S_{i,j}$	The similarity measure between users u_i and u_j based on their respective semantic transition matrices.
$\mathbf{D}$	The matrix of pairwise user dissimilarities $D_{i,j}$ used as the clustering representation.
$D_{i,j}$	The dissimilarity measure between users u_i and u_j, calculated as $D_{i,j} = 1 - S_{i,j}$, where $S_{i,j}$ is the similarity between users.
$\hat{\mathbf{D}}$	The normalized version of the user dissimilarity matrix $\mathbf{D}$.
$\mathcal{G}$	The set of user clusters formed by applying K-means clustering algorithm to the normalized distance matrix $\hat{\mathbf{D}}$.
τ	Privacy protection threshold for selecting semantic leaf nodes into $\mathcal{C}_1$.
θ, δ, γ	Direction deviation bound, geographic distance bound, and minimum inter-candidate separation used to build $\mathcal{C}_2$.
K, N_g	Anonymity set size (1 true point plus $K-1$ pseudo-locations) and number of lifestyle groups, respectively.

$$P(j) = \frac{c_j}{\sum c_j}. \tag{1}$$

c_j denotes the number of queries associated with semantic class j in the stay point trajectory. The resulting semantic cell entropy H_S is a marginal, or unconditioned, entropy. It describes the overall category distribution but does not specify the next category after a known current category c_q. Therefore, a high marginal H_S does not rule out a concentrated conditional transition distribution.

$$H_S = -\sum_j P(j) \log P(j). \tag{2}$$

The semantic transition pattern for user u is the conditional distribution of the next semantic category given the current category c_q. It describes multiple possible next semantics rather than a deterministic route. The corresponding transition counts and conditional probability are defined below.

$$P(c_{q+1} = y \mid c_q, u) = \frac{n(c_q, y)}{n(c_q)}. \tag{3}$$

Here, $n(c_q, y)$ represents the number of observed transitions from c_q to candidate next category y, and $n(c_q)$ denotes the total number of observed transitions originating from c_q. When observations are sparse, these conditional estimates are incomplete or unstable, which motivates group-level aggregation. The conditional semantic transition entropy is defined as follows.

$$H_T(c_q, u) = -\sum_y P(y \mid c_q, u) \log P(y \mid c_q, u). \tag{4}$$

The variable y denotes each candidate next semantic index. Higher entropy values correspond to more uniform conditional transition distributions and make semantic ranking more difficult. Low semantic transition entropy indicates concentration around a small number of dominant patterns, which increases predictability and allows an attacker to remove low-probability dummies with higher confidence. Thus, an unconditioned entropy or a sparsely estimated conditional entropy is insufficient when inference is conditioned on the current semantic category.

Top-N location de-anonymization attack [18] targets the spatial distribution and hierarchical clustering patterns of locations in published anonymized datasets. This attack focuses on dense location clusters to infer the user's frequently visited places or semantically significant hotspots. When these hotspots remain distinguishable after anonymization, adversaries can recover private behavioral profiles and weaken the protection offered by standard geographic obfuscation mechanisms.

C. The Overview of GeoTrace

To meet the trajectory privacy threats discussed in Section II-B, we propose GeoTrace, a geospatial-aware semantic anonymization framework for environments with uneven semantic distributions. GeoTrace emphasizes semantic transition characteristics rather than relying only on geographic proximity. It integrates group-level semantic knowledge, spatial feasibility constraints, and entropy-guided dummy generation to mitigate inference risks induced by semantic imbalance.

In the assumed setting, mobile users report stay-point trajectories through edge-assisted services to an LBS provider, and the adversary observes the published K-anonymous sets together with historical query statistics and public geographic knowledge. GeoTrace separates offline preparation from online anonymization. Offline, stay points are extracted, users are grouped, and group semantic trees are built. Online, each query point is anonymized by retrieving candidates from the requester's group structures and applying entropy-guided selection under the spatial and direction constraints. Pseudo-locations are therefore drawn from observed group support rather than from an unconstrained geographic space. Fig. 2 shows the three-stage workflow, with each component detailed in Sections II-D–II-F.

D. Semantic Data Preparation

The semantic data preparation stage standardizes raw trajectory records to support the GeoTrace framework. This stage processes raw mobility data to identify privacy-relevant stay points, partition users into lifestyle groups, and assign Points of Interest (POIs) according to group semantic characteristics. The procedure includes stay point extraction, lifestyle group division, and TF-IDF-based semantic assignment.

Authorized licensed use limited to: Tiangong University. Downloaded on October 12,2026 at 10:58:42 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Internet of Things Journal. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JIOT.2026.3741571

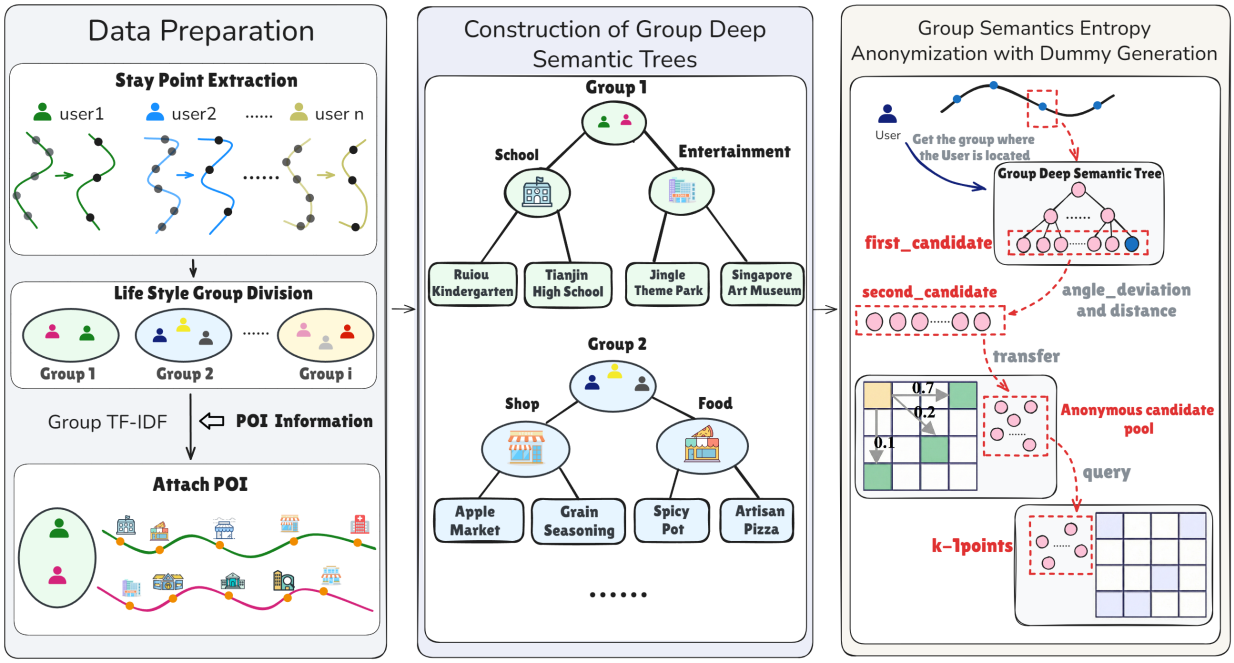


Fig. 2. The overview of GeoTrace.

Within raw trajectory data, check-in points exhibit different levels of privacy sensitivity. Not all locations require protection. The analysis therefore concentrates on stay points [8], [11], which represent locations associated with meaningful user presence. Stay points are classified into two categories. Regular stay points usually correspond to semantically stable user behaviors, such as residence or workplace visits, whereas hovering areas frequently appear in commercial or transportation regions with dense mobility interactions. These hovering regions exhibit stronger semantic ambiguity and therefore require group-level semantic assignment to avoid instability caused by sparse individual observations. Regular stay points satisfy a temporal threshold θ_t of 10 minutes and a spatial threshold θ_d of 0.1 km, following the stay-point thresholds adopted in [9], so a 10-minute stay within 100 m is treated as one regular stay point. Hovering areas satisfy the same temporal threshold but adopt a spatial threshold of 1 km, so a 10-minute stay within 1 km is treated as one hovering region. The two scales serve different purposes: 100 m is sufficient to identify fixed, small places such as residences and offices, whereas large venues such as shopping malls and transportation hubs span a wider area that a 100 m radius cannot cover. The 1 km scale is chosen to cover the spatial extent of such large venues, so that a visit within one facility is not fragmented into multiple hovering regions. For regular stay points, POI information follows the nearest semantic association rule. Hovering areas rely on group partitioning and TF-IDF-based semantic weighting to preserve semantic plausibility under privacy constraints.

Lifestyle group division operates on semantic transition statistics rather than purely spatiotemporal proximity [19]. Following Algorithm 1, each user trajectory is first represented as a generalized semantic sequence C_u over semantic categories. Transition counts between adjacent elements are accumulated in a transition matrix $\mathbf{M_C}$. These counts are aggregated into an overall matrix $\mathbf{M_P}$, from which pairwise user similarity scores $S_{i,j}$, dissimilarity $D_{i,j} = 1 - S_{i,j}$, and the normalized matrix $\hat{\mathbf{D}}$ are derived. K-means clustering over $\hat{\mathbf{D}}$ then partitions the m users into N_g lifestyle groups $\mathcal{G} = \{G_1, \ldots, G_{N_g}\}$ (N_g denotes the group count and is distinct from the anonymity set size K). This grouping strategy preserves behavioral consistency while mitigating sparsity effects caused by individual trajectory variability. Users sharing similar semantic transition patterns often exhibit comparable mobility routines and semantic preferences. Group-level aggregation therefore allows GeoTrace to introduce semantically plausible candidate locations beyond individual trajectory observations, thereby enlarging the feasible anonymization space.

Semantic assignment for hovering areas integrates lifestyle grouping with the TF-IDF weighting scheme [20]. The TF-IDF mechanism quantifies the relative importance of semantic attributes within group contexts. The importance weight of semantic i is defined as

$$W_i = \frac{m_i}{\sum m_i} \times \log \frac{M_i}{\sum M_i}. \tag{5}$$

Here, W_i denotes the importance of semantic i at the stay point, m_i represents the number of POI instances associated with semantic i within the stay region, and M_i denotes the total occurrences of semantic i within the corresponding lifestyle group. The semantic label with the highest weight is assigned to the stay point. This strategy maintains semantic coherence while incorporating both individual and group-level information.

User clustering converts per-user semantic sequences into group assignments: transition counts feed the similarity computation, and the resulting lifestyle groups index the group semantic trees built next. The group labels produced here determine which shared semantic support each user draws on during anonymization.

Authorized licensed use limited to: Tiangong University. Downloaded on October 12,2026 at 10:58:42 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Internet of Things Journal. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JIOT.2026.3741571

Algorithm 1 User Clustering Algorithm

Input:
$\mathcal{U} = \{u_1, u_2, \ldots, u_m\}$: Set of m users
$C = \{C_1, C_2, \ldots, C_m\}$: Collection of generalized semantic sequences, where $C_u = \{c_{u,1}, c_{u,2}, \ldots, c_{u,n_u}\}$ is the sequence for user u
n_u: Length of the generalized semantic sequence for user u
Output:
$\mathcal{G} = \{G_1, G_2, \ldots, G_{N_g}\}$: Set of N_g clusters of users
procedure USER CLUSTERING
  for each user $u \in \mathcal{U}$ **do**
    Initialize $\mathbf{M_C}(u)$ to zero over the common semantic-category alphabet
    for $v = 1$ to $n_u - 1$ **do**
      $M_C(u)[c_{u,v}, c_{u,v+1}] \leftarrow M_C(u)[c_{u,v}, c_{u,v+1}] + 1$
    end for
  end for
  Compute the overall transition matrix $\mathbf{M_P} = \sum_{u \in \mathcal{U}} \mathbf{M_C}(u)$
  Compute the similarity matrix $S_{i,j} = \frac{\mathbf{M_P}(i,j)}{\|\mathbf{M_P}(i)\|_1 \|\mathbf{M_P}(j)\|_1}$
  Compute the dissimilarity matrix $D_{i,j} = 1 - S_{i,j}$
  Normalize the dissimilarity matrix $\hat{\mathbf{D}} = \frac{\mathbf{D} - \min(\mathbf{D})}{\max(\mathbf{D}) - \min(\mathbf{D})}$
  Apply K-means clustering to partition users into N_g clusters $\mathcal{G}$
end procedure

The semantic data preparation stage produces trajectory representations that integrate individual semantics with group behavioral structures. These representations provide consistent inputs for subsequent semantic tree construction and entropy-based anonymization.

E. Group Semantic Tree Construction

Existing studies [9] primarily construct personal deep semantic trees from stay point data derived from individual trajectories. These structures depend strongly on trajectory density and semantic coverage. When trajectory observations remain limited, personal semantic trees often contain insufficient semantic branches and shallow transition structures. Such sparsity [21], [22] severely restricts the diversity of candidate semantic nodes and increases the risk that anonymization candidates concentrate around dominant semantic categories, thereby reducing representational stability. These effects become more significant under uneven semantic distributions, where individual trajectories may exhibit substantial sparsity.

To alleviate these limitations, GeoTrace introduces group semantic trees derived from lifestyle-based user partitioning. Let $\mathcal{G}$ denote the set of user groups obtained from semantic transition similarity. For each group $\mathcal{G}_i \in \mathcal{G}$, a group semantic tree $\mathcal{T}_{\mathcal{G}_i}$ aggregates semantic knowledge from its member users. Each user trajectory L_s^u serves as the input for constructing a personal semantic tree $\mathcal{T}_{P_u}$, which encodes the semantic structure associated with that user's stay points. Group trees therefore operate as shared semantic representations that mitigate variability induced by individual trajectory sparsity. Aggregating semantic labels from users with similar mobility behaviors can expand the candidate vocabulary beyond the labels observed for one individual.

Algorithm 2 Group Deep Semantic Tree Construction

Input: Trajectory information $\{L_s^u\}$, Group Partition $\mathcal{G}$
Output: Group Semantic Tree $\mathcal{T}_G$
Initialize empty Group Semantic Tree $\mathcal{T}_G$
for each group $\mathcal{G}_i$ in $\mathcal{G}$ **do**
  Initialize empty group semantic tree $\mathcal{T}_{\mathcal{G}_i}$
  for each user $u \in \mathcal{G}_i$ **do**
    Generate Personal Semantic Tree $\mathcal{T}_{P_u} \leftarrow$ GeneratePersonalSemanticTree(u, L_s^u)
    for each node N in $\mathcal{T}_{P_u}$ **do**
      if N is not in $\mathcal{T}_{\mathcal{G}_i}$ **then**
        Add node N to $\mathcal{T}_{\mathcal{G}_i}$
      end if
    end for
  end for
  Add $\mathcal{T}_{\mathcal{G}_i}$ to $\mathcal{T}_G$
end for
return Group Semantic Tree $\mathcal{T}_G$

In this construction, each node N is a semantic label drawn from a member's personal tree $\mathcal{T}_{P_u}$, and the membership test keeps the group tree as the union of distinct member labels without duplicating shared semantics. The procedure as written aggregates label sets. It does not merge transition counts or attach separate location indexes at this step. Locations enter later, when leaf semantics in $\mathcal{T}_G$ are mapped to trajectory points during candidate construction. The constructed group semantic trees provide stabilized semantic structures that support subsequent entropy-based anonymization.

F. Semantic Entropy Anonymization

Urban mobility scenarios frequently exhibit dense yet uneven distributions of Points of Interest. Semantic imbalance alters transition regularities, while inference-oriented attacks exploit these distortions to filter pseudo-locations. Privacy preservation therefore requires mechanisms that regulate semantic uncertainty.

GeoTrace addresses these vulnerabilities through a semantic entropy anonymization mechanism that integrates group semantic structures with entropy-based selection criteria [23]. Let $\mathcal{T}_G$ denote the group semantic tree constructed from lifestyle-based aggregation. The anonymization process operates on two candidate sets. The first candidate set, $\mathcal{C}_1$, stores semantic leaf nodes selected according to a privacy protection threshold τ. The second candidate set, $\mathcal{C}_2$, contains trajectory points that satisfy semantic and geographic plausibility constraints. These sets define the feasible space for dummy location generation.

Algorithm 3 formalizes the anonymization procedure. The algorithm first initializes the semantic candidate set $\mathcal{C}_1$ by traversing $\mathcal{T}_G$ and selecting leaf nodes that meet the threshold τ. This constraint ensures that only semantics associated with

Authorized licensed use limited to: Tiangong University. Downloaded on October 12,2026 at 10:58:42 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Internet of Things Journal. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JIOT.2026.3741571

sufficient uncertainty contribute to anonymization. For each trajectory point requiring protection, a second candidate set $\mathcal{C}_2$ is constructed. Candidate points must satisfy two conditions. The deviation between actual and predicted movement directions remains bounded by θ, and the geographic distance from the true location remains within δ. These constraints preserve trajectory realism while preventing implausible spatial transitions. The adversary considered here observes the published anonymity set together with historical query statistics and public geographic knowledge. Such an adversary may filter candidates along three lines, geographic plausibility (whether a candidate lies in a reachable, sensible area), movement consistency (whether consecutive candidates and timestamps imply abrupt speed or direction changes), and semantic likelihood (whether a candidate's semantic category is probable given the current category). Without such constraints, pseudo-locations may produce unrealistic movement trajectories or semantically implausible transitions, which may be filtered when they conflict with the adversary's available geographic or transition knowledge. To avoid structural clustering, each candidate point in $\mathcal{C}_2$ also maintains a minimum separation distance γ from other candidates.

Algorithm 3 Generation of Group Semantics Entropy Anonymization with Distance Thresholding

Input: Trajectory information, Privacy protection level τ
Output: K-anonymous set

Retrieve the requester's lifestyle group G_i and its tree $\mathcal{T}_{G_i}$
Initialize $\mathcal{C}_1$ with leaf semantics in $\mathcal{T}_{G_i}$ meeting privacy level τ
for each trajectory point p needing anonymization **do**
    Initialize empty second candidate set $\mathcal{C}_2$
    for each candidate point $p_i \neq p$ associated with a semantic in $\mathcal{C}_1$ **do**
        if Deviation between actual and predicted movement direction $< \theta$ and geographical distance $< \delta$ **then**
            Add p_i to $\mathcal{C}_2$
        end if
    end for
    Filter $\mathcal{C}_2$ such that for each point $p_j \in \mathcal{C}_2$, distance from p_j to all other points in $\mathcal{C}_2$ is greater than γ
    Select top 4K points with highest semantic transition entropy from $\mathcal{C}_2$
    Form anonymous candidate pool $\mathcal{C}_A$
    Select $K-1$ distinct points with highest cell entropy from $\mathcal{C}_A$
    Output these points together with the true point p as its K-anonymous set
end for

From $\mathcal{C}_2$, the algorithm selects the top 4K points ranked by semantic transition entropy to form the anonymous candidate pool $\mathcal{C}_A$. Thus $\mathcal{C}_1$ restricts which semantics may appear, $\mathcal{C}_2$ restricts which positions are admissible for the point being protected, and $\mathcal{C}_A$ keeps the 4K admissible positions with the most uniform conditional transition distributions. This entropy-guided ranking is intended to favor candidates with dispersed conditional distributions. The final K-anonymous set is obtained by selecting the top $K-1$ points ranked by semantic cell entropy. This selection strategy regulates semantic predictability while maintaining consistency with group behavioral structures. These constraints are intended to limit implausible candidates. Their evaluated effects are discussed in the experiments.

III. Experiments

A. Dataset

We use the Geolife dataset [24], which contains GPS trajectory records from 182 users collected over five years (comprising 17,621 trajectories). The analysis restricts the data to trajectories within Beijing's sixth ring road to reflect urban mobility scenarios. POI data from Baidu Maps [25] is incorporated to provide semantic information. During the data preprocessing, stay points are defined as locations where consecutive check-ins remain within 100 meters and the residence time exceeds 10 minutes. The final dataset contains 200 candidate trajectories selected from 20 users with the highest numbers of stay points, with each user contributing 10 trajectories. Ten repeated experiments per user are conducted to evaluate privacy preservation and data utility. The selected trajectories exhibit dynamic mobility patterns and uneven urban semantic distributions relevant to edge-assisted intelligent transportation scenarios. We set four Research Questions (RQs) to evaluate the performance of GeoTrace.

B. Baselines

MSP [9] constructs anonymous location sets using personal semantic trees, which emphasizes semantic consistency by incorporating semantic relationships between users and locations. **RDG** [11] integrates static geographic query probabilities with dynamic transition probabilities between geographic blocks. It prioritizes high-probability regions to preserve geographic continuity within anonymized sets.

C. Evaluation Metrics

Transfer Direction Deviation (DD). The DD metric measures deviations in location transfer directions. It is defined as the cosine of the angle θ^i formed by the synthetic location a, the original location c, and the previous location b,

$$\theta^i = \angle abc, \qquad DD^i = \cos(\theta^i). \tag{6}$$

Larger average cosine values indicate smaller directional distortions. DD therefore measures trajectory directional consistency as a geometric utility indicator.

Average Identification Possibility (AIP). AIP evaluates privacy exposure under single-point attacks. The metric is defined as

$$AIP = \frac{1}{(K - K')}, \tag{7}$$

where K denotes the number of points in the anonymity set, and K' represents identifiable points. Here K' counts the candidates excluded under the assumed filtering rule. The metric assumes a uniform guess among the remaining $K - K'$ candidates. Smaller AIP values imply stronger protection.

Authorized licensed use limited to: Tiangong University. Downloaded on October 12,2026 at 10:58:42 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Internet of Things Journal. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JIOT.2026.3741571

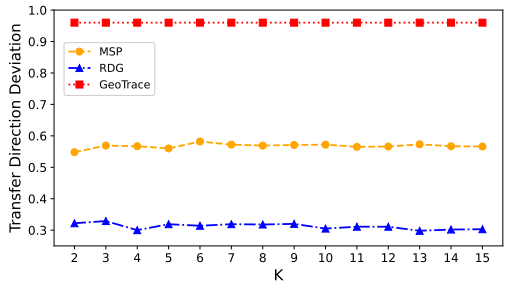


Fig. 3. Comparison of DD for different protection methods.

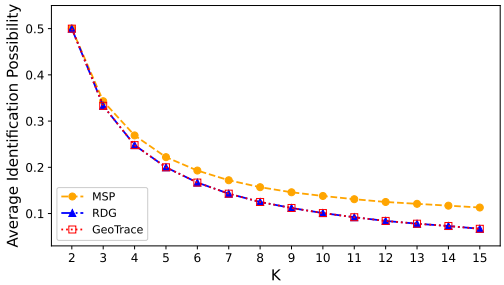


Fig. 4. Comparison of AIP for different protection methods.

Semantic Transfer Entropy. This metric quantifies the uncertainty faced by an attacker during semantic inference. It is computed as

$$\text{Semantic Transfer Entropy} = -\sum_{i} P(f, a_i) \log P(f, a_i). \tag{8}$$

Larger entropy values indicate greater uncertainty in the distribution being measured.

Top-n Position Obfuscation Attack Success Rate. The attack success rate measures the frequency of successful trajectory inference under Top-n attacks. A trajectory is considered compromised once any position is correctly identified. It is shown as

$$\text{ASR} = \frac{N_{\text{success}}}{N_{\text{total}}}, \tag{9}$$

where N_{success} denotes successfully inferred trajectories, and N_{total} represents the total number of evaluated trajectories.

D. Does GeoTrace Preserve Trajectory Direction Consistency (RQ1)

This RQ examines whether GeoTrace preserves trajectory direction consistency. The DD metric (Section III-C) quantifies angular distortion between true and synthetic transitions. Larger values indicate stronger directional alignment and better trajectory realism. Fig. 3 presents the DD values under different K settings. GeoTrace remains above 0.9 across the evaluated K values, while MSP fluctuates around 0.55–0.58 and RDG remains around 0.30–0.33. These results show that GeoTrace preserves trajectory direction more effectively than MSP and RDG. The higher DD values are consistent with the direction-constrained candidate selection in GeoTrace.

E. How Does GeoTrace Affect Single-Point Identification Risk (RQ2)

We investigate whether GeoTrace reduces single-point identification risk under different anonymity levels. Single-point exposure represents a fundamental privacy threat in trajectory anonymization, as attackers may filter implausible dummy locations through auxiliary information. The Average Identification Possibility (AIP) metric quantifies the resulting identification risk.

Fig. 4 illustrates the AIP values across varying K settings. For GeoTrace, AIP is approximately 0.5 when $K = 2$ and steadily decreases as K increases, stabilizing around 0.1 once $K > 10$. RDG exhibits a closely overlapping declining pattern and reaches approximately 0.07 at $K = 15$, while MSP remains higher at approximately 0.11. These results show that GeoTrace maintains identification risk close to RDG and lower than MSP across the evaluated anonymity levels.

This reduction is related to the candidate selection mechanism in GeoTrace. The semantic, spatial, and directional constraints reduce the inclusion of implausible dummy locations that can be readily excluded by the adversary. Consequently, more candidates remain after filtering, which decreases the uniform-guess identification possibility for a fixed K. The results therefore indicate that GeoTrace reduces single-point identification risk while maintaining plausible candidate locations under different anonymity levels.

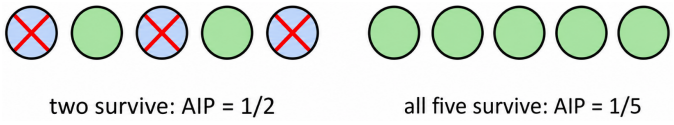


Fig. 5. The AIP measure for $K = 5$. The value is $1/2$ after three implausible candidates are excluded and $1/5$ when all five candidates are retained.

F. Can Semantic Uncertainty Mitigate Transition-Based Inference (RQ3)

Semantic Transfer Entropy characterizes uncertainty in the evaluated transition distribution. Higher Semantic Transfer En-

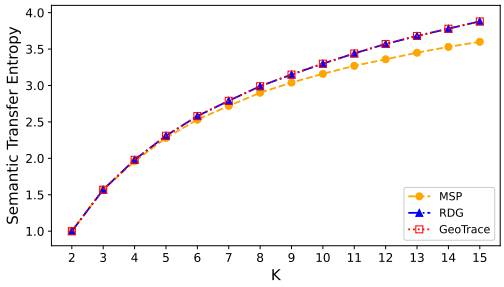


Fig. 6. Comparison of semantic transfer entropy for different protection methods.

Authorized licensed use limited to: Tiangong University. Downloaded on October 12,2026 at 10:58:42 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Internet of Things Journal. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JIOT.2026.3741571

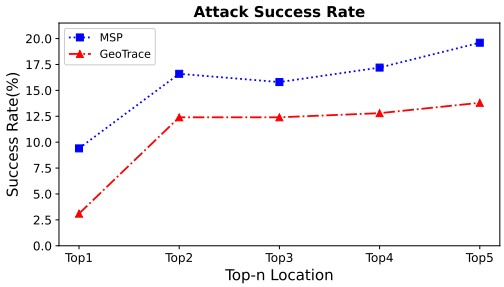


Fig. 7. K=10 Top-n position obfuscation attack success rate for different protection methods.

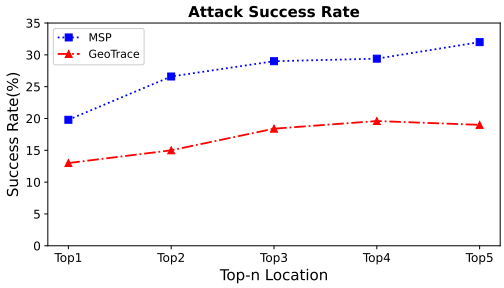


Fig. 8. K=15 Top-n position obfuscation attack success rate for different protection methods.

tropy reflects a more dispersed semantic transition distribution and increases the uncertainty of transition-based inference. Fig. 6 presents Semantic Transfer Entropy under different K settings. GeoTrace and RDG exhibit closely overlapping curves at the available plotting resolution, whereas MSP generally has lower values. Entropy increases with K, including above 10. The increasing anonymity-set size also expands the support of the semantic distribution, which contributes to the growth of Semantic Transfer Entropy as K increases. Group-level semantic support and entropy-guided selection provide a possible explanation for the dispersed distributions.

G. Is GeoTrace Robust Against Top-n Inference Attacks (RQ4)

The Top-n Position Obfuscation Attack Success Rate quantifies the probability that an adversary successfully infers true trajectory positions under Top-n inference. Smaller values correspond to stronger protection effectiveness. In this setting, Top-n denotes whether the true location appears within the first n candidate positions identified by an attacker. The RDG method exhibits an attack success rate approaching 1 across all evaluated conditions, indicating that RDG fails to provide effective resistance. Consequently, the following analysis focuses on GeoTrace and MSP.

Fig. 7 reports the attack success rates for $K = 10$. The curves generally increase with n, although MSP has a local decrease from Top-2 to Top-3. Despite this increase, the values remain below 20%, indicating that both methods maintain resistance to Top-n inference. For identical n values, GeoTrace consistently achieves lower success rates than MSP, which reflects stronger robustness. Fig. 8 presents the results for $K = 15$. Success rates increase relative to $K = 10$. GeoTrace remains below 20% across the evaluated n, while MSP rises above 30% at the largest n. GeoTrace continues to outperform MSP, with a larger gap at Top-5 than at Top-1. GeoTrace achieves lower Top-n attack success rates than MSP at both $K = 10$ and $K = 15$.

H. Discussion

The experimental results show how group-level semantic information contributes to trajectory anonymization. GeoTrace constructs each group semantic tree from users with similar semantic transition patterns, which expands the semantic labels available to an individual user. This shared semantic support increases the number of plausible alternatives considered during candidate generation. Algorithm 3 further filters these alternatives according to semantic, spatial, and directional constraints before the final anonymous set is generated. The resulting candidates therefore combine semantic diversity with geographic and movement consistency. This design is consistent with the high DD values reported in RQ1 and the lower AIP values observed as K increases.

The $K = 5$ example in Fig. 5 illustrates how candidate quality affects single-point identification risk. When three implausible candidates are removed, only two possible locations remain and the AIP value becomes $1/2$. When all five candidates remain plausible, the value decreases to $1/5$. GeoTrace reduces such candidate elimination through the joint constraints applied during pseudo-location selection. The entropy results in RQ3 provide another view of this process. GeoTrace maintains dispersed semantic transition distributions across different anonymity levels, while MSP generally produces lower Semantic Transfer Entropy. Group semantic trees enlarge the available semantic support, and entropy-guided selection favors candidates with more diverse transition possibilities. These two components therefore complement the spatial constraints applied during candidate construction.

The four research questions also reveal the relationship between privacy protection and trajectory utility. GeoTrace maintains DD values above 0.9 across the evaluated K values, which indicates that the generated candidates preserve the original movement direction well. Its AIP values remain close to RDG and lower than MSP under most evaluated anonymity levels. Semantic Transfer Entropy remains comparable to RDG and generally higher than MSP, while the Top-n experiments show lower attack success rates than MSP at both $K = 10$ and $K = 15$. These results show that GeoTrace increases semantic uncertainty while preserving trajectory directional consistency. The candidate selection process contributes to this balance because semantic diversity expands the anonymity space, while spatial and directional constraints preserve trajectory plausibility.

Several extensions can further strengthen this line of research. Future experiments can examine how lifestyle-group granularity affects semantic diversity and candidate quality under different mobility distributions. Additional datasets from other cities can evaluate whether the observed transition patterns remain consistent across geographic regions. The hovering-region scale

Authorized licensed use limited to: Tiangong University. Downloaded on October 12,2026 at 10:58:42 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Internet of Things Journal. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/JIOT.2026.3741571

can also be examined under different spatial thresholds to study its influence on semantic assignment and candidate generation. Computational analysis can further quantify the cost of group-tree construction and online anonymization in edge-assisted environments. Long-term trajectory releases and diffusion-based synthetic trajectory generation also provide useful directions for extending GeoTrace beyond the current candidate-based anonymization setting.

IV. Related Work

A. Trajectory Privacy Preservation Mechanisms

Trajectory privacy preservation reduces location disclosure while retaining mobility data utility. Existing mechanisms include pseudonymization, mix zones, and dummy generation [3], [26], while k-anonymity and differential privacy provide alternative protection mechanisms [27]. Pseudonymization protects direct identifiers but remains vulnerable to behavioral inference [28]. Mix zones reduce continuous tracking through pseudonym changes [29], but their reliance on pseudonym management and trusted coordination limits edge-side deployment [30]. Recent differential-privacy methods protect different trajectory representations. TraCS perturbs trajectory coordinates during collection [31], whereas PrivSTD releases aggregate spatio-temporal density data [32]. These mechanisms protect identifiers, collected coordinates, or aggregate statistics, while GeoTrace focuses on ambiguity among candidate locations within published anonymity sets.

B. Semantic-Aware Dummy Trajectory Generation

Dummy generation directly constructs alternative locations and avoids the trusted coordination required by mix zones [13], [33]. This property supports dynamic edge environments and privacy-aware mobility services [4], [5], [34]. Early methods mainly consider coordinate perturbation and geographic plausibility, while recent semantic approaches improve the realism of synthetic trajectories [35]. Individual-level semantic structures, however, remain sensitive to sparse observations and dominant semantic patterns. Related representation studies also demonstrate the importance of effective structures under limited samples and complex relations [36], [37].

Recent graph-based methods organize trajectory information through tree suppression or construct anonymity sets through bipartite matching [38], [39]. TGSA further introduces group aggregation for trajectory anonymization [10]. GeoTrace extends this direction through group deep semantic trees, entropy-guided candidate selection, and explicit spatial and directional constraints. The resulting candidate construction incorporates shared semantic support while preserving geographic and movement plausibility.

C. Inference Attacks via Probability Exploitation

Candidate plausibility alone cannot prevent inference when mobility data contain exploitable probability structures. Geographic transition attacks rank candidate locations according to historical transition likelihoods, and RDG incorporates dynamic geographic transition factors into privacy protection [11], [12]. Semantic information introduces additional inference signals because uneven POI distributions can concentrate semantic transition probabilities [40]. Ranking-based attacks exploit these distributions even when candidates satisfy geographic constraints, while infrastructure-level protection addresses network operations rather than semantic uncertainty [41]. GeoTrace therefore incorporates semantic transition entropy into pseudo-location selection to disperse conditional transition probabilities within anonymity sets.

Recent LLM-assisted methods extend trajectory privacy toward distributed preparation and synthetic generation. FedTDP combines federated trajectory preparation with privacy-preserving aggregation [42], while Geo-Llama generates synthetic trajectories under spatiotemporal constraints [43]. Generative trajectory studies further emphasize the importance of privacy units and evaluation protocols [44]. GeoTrace instead targets conditional semantic inference within candidate-based anonymity sets and regulates this uncertainty during pseudo-location generation.

V. Conclusions

Trajectory privacy preservation under semantic imbalance remains a challenging problem in urban IoT environments. This study investigates privacy risks caused by distorted semantic transition regularities and introduces the Dynamic Semantic Probability Distribution Attack. We propose GeoTrace, a geospatial-aware trajectory group semantic anonymization framework that combines group semantic structures with entropy-guided pseudo-location selection and spatial constraints. Experiments on the selected Geolife trajectories show that GeoTrace maintains Transfer Direction Deviation above 0.9 across different K values and achieves lower Top-n attack success rates than MSP at $K = 10$ and $K = 15$. The Semantic Transfer Entropy results further show more dispersed semantic transition distributions under the evaluated anonymity levels. These results indicate that GeoTrace improves semantic uncertainty while preserving trajectory directional consistency. We will further examine cross-dataset generalization, computational efficiency, sensitivity to spatial parameters, and diffusion-based synthetic trajectory generation.

Acknowledgment

This work is supported by the National Natural Science Foundation of China (T2600203, 62672168, 62302148).

References

[1] T. Sun, K. Fu, W. Huang, K. Zhao, Y. Gong, and M. Chen, "Going where, by whom, and at what time: Next location prediction considering user preference and temporal regularity," in *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2024, pp. 2784–2793.

[2] F. Huang, J. Xu, and J. Weng, "Multi-task travel route planning with a flexible deep learning framework," *IEEE Transactions on Intelligent Transportation Systems*, vol. 22, no. 7, pp. 3907–3918, 2021.

[3] Z. Zhang, T. Feng, W.-C. Wong, and B. Sikdar, "A geo-indistinguishable context-based mix strategy for trajectory protection in vanets," *IEEE Transactions on Vehicular Technology*, vol. 72, no. 12, pp. 16 538–16 552, 2023.

Authorized licensed use limited to: Tiangong University. Downloaded on October 12,2026 at 10:58:42 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted,
but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Internet of Things Journal. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JIOT.2026.3741571

[4] J. Tang, H. Zhu, R. Lu, X. Lin, H. Li, and F. Wang, "DLP: Achieve customizable location privacy with deceptive dummy techniques in LBS applications," *IEEE Internet of Things Journal*, vol. 9, no. 9, pp. 6969–6984, 2022.

[5] H. Guo, X. Chen, X. Zhou, and J. Liu, "Trusted and efficient task offloading in vehicular edge computing networks," *IEEE Transactions on Cognitive Communications and Networking*, vol. 10, no. 6, pp. 2370–2382, 2024.

[6] H. Guo, X. Zhou, J. Wang, J. Liu, and A. Benslimane, "Intelligent task offloading and resource allocation in digital twin based aerial computing networks," *IEEE Journal on Selected Areas in Communications*, vol. 41, no. 10, pp. 3095–3110, 2023.

[7] G. Qiu, G. Tang, C. Li, L. Luo, D. Guo, and Y. Shen, "Differentiated location privacy protection in mobile communication services: A survey from the semantic perception perspective," *ACM Computing Surveys*, vol. 56, no. 3, pp. 1–36, 2024.

[8] Y. Dai, J. Shao, C. Wei, D. Zhang, and H. T. Shen, "Personalized semantic trajectory privacy preservation through trajectory reconstruction," *World Wide Web*, vol. 21, no. 4, pp. 875–914, 2018.

[9] G. Qiu, D. Guo, Y. Shen, G. Tang, and S. Chen, "Mobile semantic-aware trajectory for personalized location privacy preservation," *IEEE Internet of Things Journal*, vol. 8, no. 21, pp. 16 165–16 180, 2021.

[10] M. Dong, Z. Wang, Z. Han, Y. Bai, C. Yuan, and G. Qiu, "Tgsa: Trajectory group semantic anonymization," in *International Conference on Security and Privacy in Communication Systems*. Springer, 2024, pp. 115–136.

[11] S. Shaham, M. Ding, B. Liu, S. Dang, Z. Lin, and J. Li, "Privacy preservation in location-based services: A novel metric and attack model," *IEEE Transactions on Mobile Computing*, vol. 20, no. 10, pp. 3006–3019, 2021.

[12] B. Niu, Q. Li, X. Zhu, G. Cao, and H. Li, "Achieving k-anonymity in privacy-aware location-based services," in *IEEE INFOCOM 2014-IEEE conference on computer communications*. IEEE, 2014, pp. 754–762.

[13] Y. Li, X. Li, X. Luo, Z. Li, H. Li, and X. Zhang, "AnotherMe: A location privacy protection system based on online virtual trajectory generation," *IEEE Transactions on Dependable and Secure Computing*, vol. 21, no. 4, pp. 2552–2567, 2024.

[14] S. Siddiqie, A. Mondal, and P. K. Reddy, "An improved dummy generation approach for infeasible regions," *Applied Intelligence*, vol. 53, no. 15, pp. 18 700–18 714, 2023.

[15] Z. Zheng, Z. Li, H. Jiang, L. Y. Zhang, and D. Tu, "Semantic-aware privacy-preserving online location trajectory data sharing," *IEEE Transactions on Information Forensics and Security*, vol. 17, pp. 2256–2271, 2022.

[16] W. Liu, Z. Wang, C. Wu, Y. Bai, and J. Zhang, "Hierarchical attention network with correction for cross-domain user association," in *Proceedings of the 40th AAAI Conference on Artificial Intelligence*, vol. 40, no. 18. AAAI Press, 2026, pp. 15 351–15 359.

[17] C. Wu, Z. Wang, K. Cen, Y. Bai, and J. Hao, "Cross-domain trajectory association based on hierarchical spatiotemporal enhanced attention hypergraph," in *Proceedings of the 39th AAAI Conference on Artificial Intelligence*, vol. 39, no. 12, 2025, pp. 12 854–12 862.

[18] L. Yu, S. Zhang, Y. Meng, S. Du, Y. Chen, Y. Ren, and H. Zhu, "Privacy-preserving location-based advertising via longitudinal geo-indistinguishability," *IEEE Transactions on Mobile Computing*, vol. 23, no. 8, pp. 8256–8273, 2024.

[19] B. Wang, H. Li, W. Wang, M. Wang, Y. Jin, and Y. Xu, "PG^2 net: Personalized and group preferences guided network for next place prediction," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 8, pp. 8655–8670, 2024.

[20] Y. Bai, S. Chen, Z. Xing, and X. Li, "ArgusDroid: detecting Android malware variants by mining permission-API knowledge graph," *Science China Information Sciences*, vol. 66, no. 9, p. 192101, 2023.

[21] H. Yuan, L. Wu, L. Xu, L. Ban, H. Wang, Y. Su, and W. Meng, "SCTP: Achieving semantic correlation trajectory privacy-preserving with differential privacy," *IEEE Transactions on Vehicular Technology*, vol. 74, no. 4, pp. 5856–5870, 2025.

[22] X. Du, H. Zhu, Y. Zheng, R. Lu, F. Wang, and H. Li, "A semantic-preserving scheme to trajectory synthesis using differential privacy," *IEEE Internet of Things Journal*, vol. 10, no. 15, pp. 13 784–13 797, 2023.

[23] M. Min, H. Zhu, S. Li, H. Zhang, L. Xiao, M. Pan, and Z. Han, "Semantic adaptive geo-indistinguishability for location privacy protection in mobile networks," *IEEE Transactions on Vehicular Technology*, vol. 73, no. 6, pp. 9193–9198, 2024.

[24] Y. Zheng, L. Zhang, X. Xie, and W.-Y. Ma, "Mining interesting locations and travel sequences from GPS trajectories," in *Proceedings of the 18th international conference on World wide web*, 2009, pp. 791–800.

[25] J. Huang, H. Wang, Y. Sun, Y. Shi, Z. Huang, A. Zhuo, and S. Feng, "ERNIE-GeoL: A geography-and-language pre-trained model and its applications in Baidu maps," in *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2022, pp. 3029–3039.

[26] M. S. AlMarshoud, A. H. Al-Bayatti, and M. S. Kiraz, "Location privacy in vanets: Provably secure anonymous key exchange protocol based on self-blindable signatures," *Vehicular Communications*, vol. 36, p. 100490, 2022.

[27] A. Majeed and S. O. Hwang, "Differential privacy and k-anonymity-based privacy preserving data publishing scheme with minimal loss of statistical information," *IEEE Transactions on Computational Social Systems*, vol. 11, no. 3, pp. 3753–3765, 2024.

[28] A. Varanda, L. Santos, R. L. d. C. Costa, A. Oliveira, and C. Rabadão, "Log pseudonymization: Privacy maintenance in practice," *Journal of Information Security and Applications*, vol. 63, p. 103021, 2021.

[29] B. Palanisamy and L. Liu, "Attack-resilient mix-zones over road networks: architecture and algorithms," *IEEE Transactions on Mobile Computing*, vol. 14, no. 3, pp. 495–508, 2015.

[30] G. Xu, J. Liu, Y. Lu, X. Zeng, Y. Zhang, and X. Li, "A novel efficient MAKA protocol with desynchronization for anonymous roaming service in global mobility networks," *Journal of Network and Computer Applications*, vol. 107, pp. 83–92, 2018.

[31] Y. Zheng and Y. Hu, "TraCS: Trajectory Collection in Continuous Space under Local Differential Privacy," *Proceedings on Privacy Enhancing Technologies*, vol. 2026, no. 3, pp. 118–138, 2026.

[32] S. Ye, Y. Hu, L. Chen, Y. Wu, Z. Zhang, T. Li, and C. S. Jensen, "PrivSTD: Differentially Private Spatio-Temporal Trajectory Density Data Publication," *Proceedings of the VLDB Endowment*, vol. 19, no. 10, pp. 2713–2726, 2026.

[33] X. Li, W. Liu, H. Feng, K. Huang, Y. Hu, J. Liu, K. Ren, and Z. Qin, "Privacy enhancement via dummy points in the shuffle model," *IEEE Transactions on Dependable and Secure Computing*, vol. 21, no. 3, pp. 1001–1016, 2024.

[34] H. Guo, X. Wu, J. Liu, B. Mao, and X. Chen, "Adaptive and reliable location privacy risk sensing in internet of vehicles," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 9, pp. 12 696–12 708, 2024.

[35] M. Dong, Z. Wang, Z. Han, X. Wang, Y. Bai, X. Ye, G. Xu, and N. N. Xiong, "Dummy-trajectory synthesis: A privacy-preserving approach for semantic trajectory data in iot-based lbsn," *IEEE Internet of Things Journal*, vol. 12, no. 18, pp. 38 031–38 042, 2025.

[36] Y. Bai, Z. Xing, X. Li, Z. Feng, and D. Ma, "Unsuccessful story about few shot malware family classification and siamese network to the rescue," in *Proceedings of the ACM/IEEE 42nd international conference on software engineering*, 2020, pp. 1560–1571.

[37] Y. Zhang, G. Xu, R. Feng, X. Li, S. Chen, Z. Xing, Y. Bai, Y. Lyu, W. Gong, and X. Zhao, "Knowledge is power: a knowledge graph-based approach for mobile malware traceability analysis," *IEEE Transactions on Mobile Computing*, vol. 25, no. 8, pp. 13 196–13 212, 2026.

[38] C.-Y. Lin, "Efficient suppression algorithms for preserving trajectory privacy," *Information Sciences*, vol. 677, p. 120837, 2024.

[39] L. Kuang, W. Shi, X. Chen, J. Zhang, and H. Liao, "A location semantic privacy protection model based on spatial influence," *Scientific reports*, vol. 15, no. 1, p. 15227, 2025.

[40] L. Zhou, R. Yang, L. Ma, G. Chen, H. Zhu, L. Yang, Q. Pei, and Q. Li, "The feasibility of location anonymity: An empirical study towards a real-world location privacy protection system in takeout services," in *IEEE INFOCOM 2025-IEEE Conference on Computer Communications*. IEEE, 2025, pp. 1–10.

[41] H. Guo, X. Wu, Z. Yin, B. Mao, Y. Xun, J. Liu, and W. Chen, "Achieving multi-attribute superiority and sybil attack detection in IoV: A heuristic-based dynamic RSU deployment scheme," *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 2, pp. 2734–2746, 2025.

[42] Z. Zeng, Z. Fang, W. Shao, L. Chen, and Y. Gao, "A Unified Federated Framework for Trajectory Data Preparation via LLMs," in *International Conference on Learning Representations*, 2026.

[43] S. Li, T. Tran, H. Lin, J. Krumm, C. Shahabi, L. Zhao, K. Shafique, and L. Xiong, "Geo-Llama: Leveraging LLMs for Human Mobility Trajectory Generation with Constraints," in *2025 26th IEEE International Conference on Mobile Data Management (MDM)*. IEEE, 2025, pp. 20–31.

[44] E. Buchholz, A. Abuadbba, S. Wang, S. Nepal, and S. S. Kanhere, "SoK: Can Trajectory Generation Combine Privacy and Utility?" *Proceedings on Privacy Enhancing Technologies*, vol. 2024, no. 3, pp. 75–93, 2024.

Authorized licensed use limited to: Tiangong University. Downloaded on October 12,2026 at 10:58:42 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.